Arunabha Majumder

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Portfolio: arunabha22.github.io/arunabha-majumder

Helsing

Robotics Mechanical Engineer

Dear Helsing team,

I'd like to apply for the Robotics Mechanical Engineer role. Designing mechanisms and actuators, building them and seeing them work inside a complete robot is the work I enjoy most, and I would be proud to do it for a team that helps protect open, democratic societies.

As a research assistant at Aalborg University I designed a hybrid shoulder exoskeleton. A variable stiffness mechanism, with a worm gear that pretensions the spring, works in parallel with a 6 Nm motor, and together they give the torque needed to lift the arm. I optimised the mechanism with a genetic algorithm (diameter, spring stiffness and pretension force), made the CAD and drawings in SolidWorks, 3D printed the first parts and had the final ones manufactured. I integrated the motor, encoder and force/torque sensors and measured the stiffness on a physical test setup. It uses over 25% less motor energy and won the Best Paper Award at ISRM 2026.

I also designed and modelled a novel 5-bar planar parallel robot with variable stiffness mechanisms (VSM), estimated its stiffness with less than 10% error and quantified human arm impedance. The force estimation came from the variable stiffness actuators and the robot's Jacobian. For the arm impedance study with this robot (6 participants), I implemented an Arduino-based trigger to sync the robot's motion with Qualisys motion capture.

In my Master's at CSIR-CMERI I worked in a team of 4 electronics and mechanical engineers on a soft exosuit. I fabricated silicone-based pneumatic muscles and designed the outer PID loop, with integral anti-windup and error nonlinearisation, of a neural network-based gain-scheduled controller in MATLAB/Simulink and LabVIEW. In my Bachelor's I did the CAD, machining and assembly of an arm exoskeleton and exo glove. I also hold a Senior Diploma in Fine Arts and do graphic design (Illustrator, Photoshop, Procreate), which helps me sketch ideas and explain designs clearly in reviews.

I'm ready to move to Germany or France for the role. I would love to bring new and innovative ideas to your robots and grow with your team. My portfolio, with videos and results from my work, is at arunabha22.github.io/arunabha-majumder. Thank you for reading.

Best,
Arunabha Majumder